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Factors Associated With Unit-Specific Failure in a University-Level Statistics Course

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ABSTRACT

Failure in introductory statistics courses is commonly evaluated using overall course grades, potentially obscuring important differences across content areas. This study examines factors associated with failure across four instructional units of a General Statistics course taught at a private university in Colombia: Exploratory Data Analysis, Probability and Random Variables, Statistical Inference, and Linear Regression. Using a complete-case analytical sample of 186 undergraduate students, we estimated unit-specific logistic regression models combining exam performance and pre-exam self-perception measures. Failure rates varied substantially across units, ranging from 19% to 46%. Students from programs outside the School of Applied Sciences and Engineering consistently exhibited higher odds of failure, while academic stage, prior programming experience, and self-perceived competence showed unit-specific effects. Robustness analyses supported the stability of these findings. The results demonstrate that academic vulnerability in statistics varies across content areas and that unit-level analyses reveal patterns hidden by aggregate course outcomes, informing more targeted pedagogical interventions.

1 | Introduction

Statistics is one of the foundational subjects in contemporary higher education due to its role in developing quantitative reasoning, data interpretation, and evidence-based decision-making skills. Nevertheless, introductory statistics courses continue to pose significant challenges for both students and instructors, as reflected in high failure rates, persistent conceptual difficulties, and substantial heterogeneity in academic performance (Garfield and Ben-Zvi 2007; Garfield et al. 2008).

Several studies have identified factors associated with success and failure in statistics, including students' individual characteristics, variables related to their prior preparation, and aspects of the educational environment (Burns et al. 2020; Rabin et al. 2021; Fabbricatore et al. 2023). However, most of this research evaluates performance using aggregate course-level indicators, such as final grades or overall assessment averages. While this approach helps identify general factors associated with academic achievement, it provides limited insight into how learning difficulties evolve across the different topics that comprise a statistics course.

This limitation is particularly relevant because statistical concepts do not entail the same level of cognitive complexity nor require the same skills from students. Consequently, a single aggregate indicator may conceal important differences across thematic units and hinder the identification of the topics that pose the greatest barriers to learning. Understanding these differences is essential for designing more precise and timely pedagogical interventions.

To contribute to this discussion, the present study examines failure rates and the factors associated with performance

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across four thematic units of a General Statistics course offered at a private university in Colombia. Unlike traditional approaches that focus on overall course outcomes, this study adopts a disaggregated perspective that allows for the examination of how patterns of success and failure vary throughout the course.

The study is guided by the following research questions:

- How do failure rates vary across the different units of the course?
- Which cognitive, affective, pedagogical, and contextual factors are associated with these variations?

By examining performance at the thematic-unit level, this study seeks to provide evidence that contributes to a more precise understanding of learning difficulties in statistics and generates useful insights for the design of targeted pedagogical interventions, the early identification of at-risk students, and the continuous improvement of teaching and learning processes.

2 | Theoretical Framework

Understanding the factors that influence students' academic performance in introductory statistics courses has been a central concern in statistics education research for several decades. As statistical literacy becomes increasingly important in a data-driven society, researchers and instructors have sought to identify the factors that facilitate or hinder learning in the discipline, as well as to design strategies aimed at reducing academic failure rates and improving students' conceptual understanding. In this context, several studies have shown that performance in statistics is a multidimensional phenomenon shaped by the interaction of cognitive, affective, pedagogical, and contextual factors (Burns et al. 2020; Rabin et al. 2021; Fabbricatore et al. 2023).

Among studies aimed at identifying predictors of academic performance, the work of Burns et al. (2020) is particularly noteworthy. The authors developed a statistical model to explain the performance of engineering students enrolled in an introductory statistics course. Their results showed that objective variables, such as laboratory and homework grades, were strong predictors of final performance, whereas some attitudinal measures exhibited more limited explanatory power. These findings suggest that academic achievement depends not only on students' mathematical abilities or individual characteristics but also on pedagogical conditions and the learning environment in which instruction takes place.

2.1 | Statistical Reasoning and Conceptual Challenges

Beyond performance observed in assessments, an important line of research has examined how students construct statistical knowledge and the main cognitive difficulties they encounter during this process. In a widely cited review, Garfield and Ben-Zvi (2007) and Garfield et al. (2008) argue that many students develop procedural skills without achieving a deep conceptual understanding of fundamental statistical ideas. Concepts that may appear basic, such as the mean, variability, sampling distributions, or statistical inference, are often understood in a fragmented manner, limiting their transfer to novel situations and hindering the solution of more complex problems.

A recurring difficulty is that students often perceive data as disconnected individual observations rather than as aggregates possessing emergent properties. As a result, concepts such as variability are frequently reduced to the mechanical computation of descriptive measures without a full understanding of their role in statistical reasoning and inference. This phenomenon becomes particularly problematic in topics related to statistical inference, where students must integrate ideas of sampling, uncertainty, distributions, and variability in order to draw evidence-based conclusions.

For this reason, contemporary statistics education literature has promoted a shift from computation-centered approaches toward approaches focused on the development of statistical reasoning. From this perspective, effective learning requires students to establish connections among data, context, and uncertainty, while progressively developing statistical literacy, statistical reasoning, and statistical thinking (Garfield and Ben-Zvi 2007; Garfield et al. 2008). These goals are unlikely to be achieved through the fragmented teaching of isolated topics and instead require educational experiences that foster conceptual integration and critical reflection.

The importance of this approach has also been emphasized by Olani et al. (2011), who argue that traditional instruction based primarily on mathematical procedures is insufficient for developing lasting statistical competencies. According to these authors, students achieve better learning outcomes when instructional activities promote interpretation, argumentation, and decision-making in authentic contexts, rather than focusing solely on the application of formulas.

2.2 | Attitudes, Self-Efficacy, and Statistical Learning

In addition to conceptual challenges, learning statistics is strongly influenced by affective and motivational factors. Numerous studies have shown that attitudes toward statistics can significantly affect students' engagement with the discipline and, consequently, their academic performance (Garfield and Ben-Zvi 2007).

Social cognitive theory, proposed by Bandura (1997), identifies self-efficacy as one of the primary determinants of human behavior. In educational settings, self-efficacy refers to a student's belief in their ability to successfully perform a specific task. Numerous studies have shown that students with higher levels of self-efficacy tend to persist longer when facing difficulties, devote more time to studying, and achieve better academic outcomes (Finney and Schraw 2003; Olani et al. 2011).

In the specific context of statistics education, Dempster and McCorry (2009) found that perceived cognitive competence is one of the strongest predictors of performance on statistical

assessments. Their results suggest that the way students perceive their own ability to understand and interpret statistical concepts may be even more influential than some traditional measures of prior academic preparation.

Similarly, Olani et al. (2011) showed that statistical self-efficacy can be strengthened through effective pedagogical practices and instructor support. In particular, they found that students who perceived higher levels of support experienced greater increases in their beliefs about their own competence than those who perceived lower levels of guidance and encouragement.

Furthermore, the literature has documented that negative attitudes toward statistics often persist even among students who achieve strong academic results. Ghulami et al. (2015), Pascual et al. (2025), and Gal and Ginsburg (1994) describe a common paradox: many students recognize the professional value of statistics while simultaneously perceiving it as a difficult, intimidating, or unappealing subject. This situation may generate anxiety, reduced motivation, and resistance toward quantitative content.

To examine these affective dimensions, researchers have frequently employed the SATS-36 scale, which assesses constructs such as *Affect*, *Cognitive Competence*, *Value*, *Difficulty*, *Interest*, and *Effort*. The psychometric robustness of this instrument has been extensively documented across a variety of educational and cultural contexts (Akour 2022).

Although perceived competence refers to students' judgments about their ability to perform specific cognitive tasks, in this study it is treated primarily as an affective-motivational construct because it reflects students' beliefs about their own capabilities rather than an objective measure of cognitive performance. We nevertheless recognize that perceived competence lies at the intersection of affective and cognitive dimensions.

2.3 | Disciplinary Differences in Statistics Learning

Disciplinary differences represent another important dimension for understanding performance in introductory statistics courses. Several studies have found that students from STEM fields often exhibit systematic advantages over students from non-quantitative disciplines due to differences in mathematical preparation, computational experience, and confidence in numerical reasoning (Rabin et al. 2021; Fabbricatore et al. 2023).

In contrast, students from psychology, education, social sciences, social work, and arts-related disciplines often perceive statistics as a major academic challenge (Gal and Ginsburg 1994). This perception may lead to statistical anxiety, reduce confidence in their own abilities, and negatively affect academic persistence.

The study by Rabin et al. (2021) found that basic mathematical skills are among the most important predictors of performance in introductory statistics courses. The authors reported that variables related to elementary mathematical knowledge explained a substantial proportion of the variation observed in students' grades. Similar findings were reported by Fabbricatore et al. (2023), who identified mathematical knowledge, self-efficacy, and attitudes toward statistics as factors significantly associated with academic performance.

Nevertheless, the literature also indicates that mathematical preparation is not the only factor that matters. Dempster and McCorry (2009) showed that students' perceptions of their own ability to learn statistics play a fundamental role in academic outcomes. Similarly, Olani et al. (2011) concluded that students with diverse mathematical backgrounds can achieve comparable levels of statistical reasoning when provided with adequate pedagogical support and engaged in learning experiences centered on conceptual understanding.

The increasing incorporation of statistical software has also partially transformed the nature of these disciplinary differences. According to Rabin et al. (2021), the use of computational tools can reduce reliance on manual calculations and promote the development of competencies related to data interpretation and evidence-based decision-making. However, these advantages do not completely eliminate differences associated with prior preparation or students' confidence in quantitative reasoning.

2.4 | From Global Course Outcomes to Topic-Level Learning

Traditionally, most studies on academic performance in statistics have relied on aggregate indicators such as final course grades or pass/fail status (Burns et al. 2020). Although these measures provide a useful assessment of overall academic success, recent research has begun to question their ability to adequately capture the learning processes that take place within a course.

In particular, Arakawa et al. (2022) argue that models based exclusively on aggregate outcomes tend to oversimplify the learning process and hinder the early identification of specific difficulties. Their findings show that academic challenges can be detected more accurately when analyses are conducted at the level of individual topics rather than relying solely on aggregated measures of performance.

Similarly, Zambrano and Baker (2024) found that knowledge and academic engagement exhibit different predictive capacities depending on the specific mathematical topic being analyzed. Their findings suggest that distinct conceptual domains may play different roles in students' educational trajectories and that patterns observed at an aggregate level can obscure important underlying relationships.

Even more direct evidence is provided by the work of Axelsen et al. (2025), who examined learning in an introductory statistics course using module-based approaches. Their results revealed that many students who passed the course nevertheless exhibited substantial conceptual gaps in specific modules. The authors describe this phenomenon as a form of fragmented learning, or *partial mastery*, in which students successfully master some topics while continuing to experience significant difficulties in others.

These findings are consistent with recent trends in learning analytics aimed at identifying specific knowledge gaps and designing more targeted pedagogical interventions. From this perspective, aggregate outcomes may conceal important patterns related to learning, motivation, and conceptual development (Arakawa et al. 2022; Zambrano and Baker 2024; Axelsen et al. 2025).

Furthermore, several studies suggest that the factors associated with academic performance may vary across specific topics. Variables such as prior mathematical preparation, computational experience, self-efficacy, attitudes toward statistics, and certain demographic characteristics do not necessarily affect all course content in the same way (Olani et al. 2011; Rabin et al. 2021; Fabbricatore et al. 2023; Gil-Hernández et al. 2024; Alharbi et al. 2016). Consequently, a disaggregated analysis by thematic units may provide a more precise understanding of learning processes and the factors that contribute to students' difficulties.

Finally, recent efforts to modernize statistics education have promoted more flexible, interdisciplinary curricula centered on the analysis of real-world data. In this vein, Burns and Hammond (2019) describe successful curriculum redesign initiatives aimed at strengthening conceptual understanding through authentic projects, applied activities, and meaningful learning experiences. These proposals reflect a growing trend toward educational models that prioritize deep understanding over the memorization of procedures.

Taken together, the available evidence suggests that learning statistics is a multidimensional process influenced by cognitive, affective, disciplinary, and pedagogical factors. It also indicates that learning difficulties may manifest differently depending on the specific content being addressed. From this perspective, analyzing student performance at the thematic-unit level provides an opportunity to better understand where and why learning difficulties arise, generating valuable evidence to inform curriculum design, assessment practices, and pedagogical interventions.

3 | Methods

3.1 | Study Design

This study employed a quantitative observational design based on institutional assessment records complemented by student survey data. The objective was to identify factors associated with failure across four thematic units of an introductory General Statistics course. Unlike studies that rely exclusively on a final course grade as a measure of performance, the present work adopted a disaggregated analytical strategy at the thematic-unit level. This methodological decision is supported by recent evidence suggesting that aggregate outcomes may conceal specific learning difficulties and heterogeneous patterns of performance across conceptually distinct topics (Arakawa et al. 2022; Zambrano and Baker 2024; Axelsen et al. 2025).

Each unit of the course addresses different competencies and constitutes a relatively autonomous conceptual domain. For this reason, unit-specific models were considered the primary analytical strategy. Nevertheless, complementary analyses based on pooled models and repeated-measures models were also conducted to assess the robustness of the findings.

3.2 | Course Context

The General Statistics course was offered during the 2024-2 academic semester at a private Colombian university. The course was delivered in three independent sections, each corresponding to a different class schedule. Students were free to enroll in the section that best fit their academic availability; therefore, assignment to sections was not random.

The three sections were taught by different instructors. Two were led by adjunct faculty members, while one was taught by a full-time faculty member. Despite this difference, all sections shared the same learning outcomes, syllabus, instructional materials, a common pool of exam questions, homogeneous assessment criteria, and standardized grading rubrics.

The course was organized into four thematic units. Unit 1, *Descriptive Statistics for a Single Variable*, focused on the organization, representation, and summarization of data. Unit 2, *Analysis of Random Phenomena*, covered concepts related to probability and random variables. Unit 3, *Statistical Inference*, focused on parameter estimation and hypothesis testing. Finally, Unit 4, *Linear Dependence Analysis*, introduced the fundamental concepts of correlation and simple linear regression.

Each unit concluded with a digitally administered examination supervised by the corresponding instructor. The exams were constructed from a shared pool of approximately 1500 questions and were administered during the same academic week, with a maximum difference of 1 day between sections.

Under institutional regulations, students were allowed to withdraw from the course through Week 15 of the semester without academic, financial, or administrative penalties. Consequently, course participation could change even after students had completed one or more unit examinations.

3.3 | Analytical Sample

The initial cohort consisted of 300 students enrolled in the General Statistics course during the 2024-2 academic semester. Of these, 254 had a recorded grade for Exam 1, and 216 had recorded grades for all four unit examinations.

To ensure that the four unit-specific models were estimated using the same set of students and complete information on the variables retained for analysis, a complete-case criterion was applied. Thirty of the 216 students who completed all four examinations were excluded because of missing information in other variables, resulting in a final analytical sample of 186 students.

To assess potential selection associated with persistence in the course, we specifically examined the subsequent participation

of students who failed Exam 1. Among the 115 students who obtained a grade below 3.0 on Exam 1, 102 (88.7%) had a recorded grade for Exam 2, and 100 (87.0%) had recorded grades for all three subsequent examinations. Ten students (8.7%) had no recorded grade in any of the subsequent examinations, representing the clearest available indicator of possible discontinuation after Exam 1. Because students could formally withdraw from the course through Week 15 without academic, financial, or administrative penalties, absence of subsequent grades should not be interpreted as a perfect measure of course withdrawal.

We also assessed whether inclusion in the final analytical sample was associated with performance on Exam 1. Among the 115 students who failed Exam 1, 29 (25.2%) were excluded from the complete-case sample, compared with 39 of the 139 students who passed Exam 1 (28.1%). There was no evidence of an association between Exam 1 pass/fail status and inclusion in the analytical sample ($\chi^2(1) = 0.13$, $p = 0.714$).

Using a common analytical sample avoids changes in sample size across the four unit-specific models and facilitates direct comparisons of the factors associated with failure. However, this approach does not eliminate potential selection associated with course persistence or complete-case inclusion. Accordingly, the findings should be interpreted as conditional on membership in the analytical sample and not necessarily as representative of all students initially enrolled in the course.

The course is offered primarily to students from the School of Applied Sciences and Engineering (ECAeI), although it is also part of the curriculum of the Interactive Design program, which is housed within the School of Arts and Humanities. Due to curricular differences across programs, students take the course at different stages of their academic trajectory. Approximately 74% of the sample was enrolled between the fourth and sixth semesters, while 58% were in their fifth semester or earlier. Students from the Interactive Design program represented 24% of the sample.

3.4 | Outcome Variable

The dependent variable was pass/fail status on each unit examination. According to institutional regulations, a student was classified as having failed when their score was below 3.0 on a grading scale ranging from 0 to 5. No grading curves or extraordinary grade adjustment mechanisms were applied.

The observed failure rates were 46% for Unit 1, 19% for Unit 2, 40% for Unit 3, and 30% for Unit 4 (Table 1).

Although a subsequent descriptive analysis was conducted on the specific components of each examination to identify particular conceptual difficulties, the regression models used only the overall pass/fail status of the complete examination as the outcome variable. Therefore, failing a specific component does not necessarily imply failing the examination as a whole.

TABLE 1 | Observed failure rates by unit.

Unit	Failure rate
Unit 1: Descriptive statistics	46%
Unit 2: Random phenomena	19%
Unit 3: Statistical inference	40%
Unit 4: Linear dependence	30%

3.5 | Explanatory Variables

The explanatory variables included demographic characteristics, academic background variables, course-section characteristics, and self-perception measures.

The demographic variables considered were Age, Sex, type of secondary school attended, and geographic origin. Academic background variables included prior experience in statistics, programming experience, stage within the academic program, and receipt of institutional financial support.

In addition, before each examination, students completed a brief survey designed to assess their perceptions of the learning outcomes associated with the corresponding unit. Responses were recorded using Likert-type scales and subsequently rescaled to the $[0, 1]$ interval, where higher values indicate a more favorable perception of their own competencies. For ease of interpretation, the effects of the self-perception variables are interpreted in terms of 0.1-unit increases on the rescaled scale; thus, the corresponding odds ratios represent the change in the odds of failure associated with a 0.1-unit increase in perceived competence.

The survey questions were specific to each thematic unit. For example, Unit 1 assessed students' perceptions of their ability to recognize the nature of data, summarize information using tables and graphs, and interpret descriptive measures. Unit 2 included questions related to probability distributions, random variables, and fundamental probability theorems. In Unit 3, the questions addressed sampling distributions, estimation, confidence intervals, and hypothesis testing. Finally, Unit 4 assessed perceived competence in interpreting linear relationships, testing regression coefficients, and interpreting goodness-of-fit measures. The complete wording of all survey items is provided in the Supplementary Material.

For the conceptual interpretation of the results, the predictors were organized according to the four dimensions introduced in the theoretical framework. Prior experience in statistics and programming, as well as stage within the academic program, were considered indicators of cognitive and academic preparation. Pre-examination self-perception measures primarily represented the affective dimension. Course-section characteristics and examination-related conditions were associated with the pedagogical dimension, whereas academic program and demographic and background characteristics were considered contextual factors. These categories are intended as an organizational framework rather than as mutually exclusive dimensions, since some constructs, particularly perceived competence, may span more than one dimension.

3.6 | Statistical Analysis

Independent binary logistic regression models were estimated for each of the four thematic units. The dependent variable was failure status (1 = failed, 0 = passed).

The unit-level modeling strategy was adopted because each assessment measures conceptually distinct competencies and because the primary objective of the study is to identify factors associated with specific learning difficulties rather than to predict an overall course outcome. This approach makes it possible to detect patterns that would remain hidden under an aggregate measure of performance.

For each unit, an initially broad set of explanatory variables was considered, including academic background, demographic characteristics, examination duration, course section, and self-perception measures. Final model selection was conducted through deviance comparisons between full and reduced specifications. In all cases, the full models provided a superior fit and were therefore retained for interpretation.

Results are reported as odds ratios, 95% confidence intervals, and statistical significance values. For the self-perception variables, which were rescaled to the $[0, 1]$ interval, substantive interpretation of the effects is based on 0.1-unit increases. This scale of change provides a more informative interpretation than that corresponding to a full one-unit increase.

Because four independent models and multiple predictors were estimated, interpretation focused primarily on the consistency of effects, the magnitude of the odds ratios, and the associated confidence intervals. Results with significance levels between 0.05 and 0.10 are considered exploratory and are interpreted with caution.

3.7 | Robustness Analyses

Additional analyses were conducted to assess whether the conclusions obtained from the unit-specific models depended on the structure of the data or on potential effects associated with course sections.

First, we evaluated the potential within-student dependence arising from the fact that each participant contributed up to four observations, one for each thematic unit. To do so, the dataset was restructured into long format, yielding 744 observations corresponding to 186 students observed across the 4 units. A logistic model with a student-level random intercept was then estimated using this structure.

The null model produced an intraclass correlation coefficient (ICC) of 0.073, indicating modest within-student dependence prior to the inclusion of covariates. However, after incorporating the observed predictors, the variance associated with the random intercept converged to zero ($\sigma_u^2 = 0$), and a likelihood ratio test indicated that the multilevel specification did not improve model fit relative to a conventional logistic regression ($\chi^2(1) = 0.000$, $p = 0.500$). These results suggest that between-student variability is largely explained by the observed covariates and that the residual dependence among observations from the same student is limited.

Second, a pooled model was estimated incorporating indicators for thematic unit, academic program, pre-examination self-perception, stage within the academic program, and performance on the immediately preceding assessment. Because prior assessment performance is undefined for Unit 1, this analysis was restricted to Units 2–4, yielding 558 observations from the 186 students. This analysis addressed a concern raised by the reviewers regarding the potential influence of prior performance on responses to the survey administered before each examination.

The results showed that self-perception remained a statistically significant predictor of failure. A 0.1-unit increase in self-perception was associated with an approximately 15.6% reduction in the odds of failure ($OR_{0.1} = 0.844$; 95% CI [0.768, 0.928]; $p < 0.001$), whereas performance on the previous assessment did not reach statistical significance (OR = 1.402; 95% CI [0.913, 2.152]; $p = 0.122$). These findings suggest that self-perception captures meaningful information about students' perceived preparedness and is not fully explained by performance on the preceding assessment.

Subsequently, we evaluated whether the effects associated with academic program varied systematically across thematic units through a likelihood ratio test comparing pooled models with and without Unit × Program interactions. The inclusion of these interactions did not result in a significant improvement in model fit ($p = 0.711$). Although this result should be interpreted cautiously given the sample size available to detect interaction effects, it provides additional evidence that the main patterns identified are relatively stable across units.

Finally, we examined the potential influence of course sections and instructors. To this end, all models were re-estimated including an indicator variable corresponding to the section taught by the full-time faculty member (Grupo_B). The inclusion of this variable produced only minor changes in the odds ratios of the main predictors and did not alter the substantive conclusions of the models. Furthermore, the effect of Grupo_B was statistically significant only in Unit 1, whereas no significant differences between sections were observed in the remaining units. Detailed estimates from these analyses are presented in the Supplementary Material.

Taken together, the robustness analyses support the validity of the unit-specific modeling strategy. The results suggest that the conclusions are not attributable to within-student dependence, clustering effects associated with course sections or instructors, or the potential influence of prior performance on self-perception responses.

4 | Results

4.1 | Overview of the Sample and Failure Rates by Unit

The analytical sample consisted of 186 students with complete information on the variables retained for analysis. These students were a subset of the 216 students who had recorded grades for all four unit examinations. As described in Section 3.3, the

same analytical sample was used across the four unit-specific models to facilitate comparisons while acknowledging the potential for selection associated with course persistence and complete-case inclusion.

Failure rates varied substantially across the four thematic units. Unit 1, corresponding to *Descriptive Statistics for a Single Variable*, exhibited the highest failure rate, at 46%. Unit 2, focused on *Analysis of Random Phenomena*, had the lowest failure rate, at 19%. Unit 3, devoted to *Statistical Inference*, showed a failure rate of 40%, while Unit 4, covering *Linear Dependence Analysis* and regression, had a failure rate of 30%.

A review of the internal components of the assessments revealed that difficulties were not distributed uniformly within each unit. In Unit 1, 36.02% of students correctly completed the task involving tabular data representation, whereas only 23.66% successfully solved items requiring the combined application of measures of central tendency, position, and dispersion. In Unit 3, 39.25% correctly solved problems related to the sampling distribution of the variance; 21.51% correctly answered general questions on confidence intervals; 43.55% succeeded on confidence intervals specifically for the variance; and only 6.45% correctly solved an integrative hypothesis-testing problem. When this latter component was disaggregated, 30.65% answered correctly questions involving hypothesis tests for the mean, 47.85% for proportions, and 31.72% for variance. In Unit 4, 47.31% of students correctly solved exercises related to linear regression and interpretation of the correlation coefficient.

These descriptive results indicate that a single aggregate measure of performance does not allow for the precise identification of the topics in which students experience the greatest difficulties.

TABLE 2 | Main predictors associated with failure by unit.

Unit	Predictor	OR	95% CI
U1	Program outside ECAeI (No_ECAeI)	3.41	1.52–7.62
U1	Full-time faculty section (Grupo_B)	3.39	1.01–11.37
U1	Recognizing the nature of data (Perp1_P1), per 0.1-unit increase	0.764	0.575–1.014
U2	Age	1.26	1.06–1.49
U2	No prior programming experience (exP)	6.42	1.37–30.09
U2	Sex	0.346	0.129–0.930
U2	Program outside ECAeI (No_ECAeI)	3.16	1.12–8.96
U2	Properties of random variables (Perp5_P2), per 0.1-unit increase	0.687	0.465–1.014
U3	Program outside ECAeI (No_ECAeI)	2.32	1.01–5.32
U3	Early stage in the academic program (etapa_sem)	2.58	0.913–7.30
U4	Program outside ECAeI (No_ECAeI)	2.98	1.14–7.82
U4	Examination duration (dur4)	0.971	0.943–1.000
U4	Testing regression coefficients (Perp2_P4), per 0.1-unit increase	0.521	0.322–0.844

Note: For self-perception variables, odds ratios and 95% confidence intervals are reported for a 0.1-unit increase on the rescaled $[0, 1]$ scale.

4.2 | Results of the Unit-Specific Logistic Models

Independent binary logistic regression models were estimated for each thematic unit. The dependent variable was failure on the corresponding examination, coded as 1 if the student failed and 0 if the student passed. The complete models, including all coefficients, standard errors, *p*-values, and odds ratios, are provided in Supplementary Material: Appendix A.

Table 2 summarizes the main predictors associated with failure in each unit. To facilitate interpretation, the functional variable names are accompanied by descriptive labels.

4.2.1 | Unit 1: Descriptive Statistics for a Single Variable

In Unit 1, the most important predictor was enrollment in a program outside ECAeI. Students from these programs were 3.41 times more likely to fail than students enrolled in engineering programs (OR = 3.41, 95% CI: 1.52–7.62). The variable corresponding to the section taught by the full-time faculty member also showed evidence of an association with failure in this unit (OR = 3.39, 95% CI: 1.01–11.37). However, this estimate is based on only 17 students in that section and has a wide confidence interval. Therefore, although the results suggest higher odds of failure in this section, the exact magnitude of the association is imprecise and should be interpreted with caution.

The self-perception variable related to the ability to recognize the nature of data (Perp1_P1) showed a marginal protective effect.

A 0.1-unit increase in self-perception was associated with an approximately 23.6% reduction in the odds of failure ($OR_{0.1} = 0.764$, 95% CI: 0.575–1.014). However, the confidence interval includes 1, so this result should be interpreted with caution. See Figure 1.

4.2.2 | Unit 2: Analysis of Random Phenomena

In Unit 2, the absence of prior programming experience had the largest point estimate among the risk factors (OR = 6.42, 95% CI: 1.37–30.09). However, this estimate is based on only 12 students without prior programming experience and has a considerably wide confidence interval. Consequently, the results suggest higher odds of failure for these students, but the exact

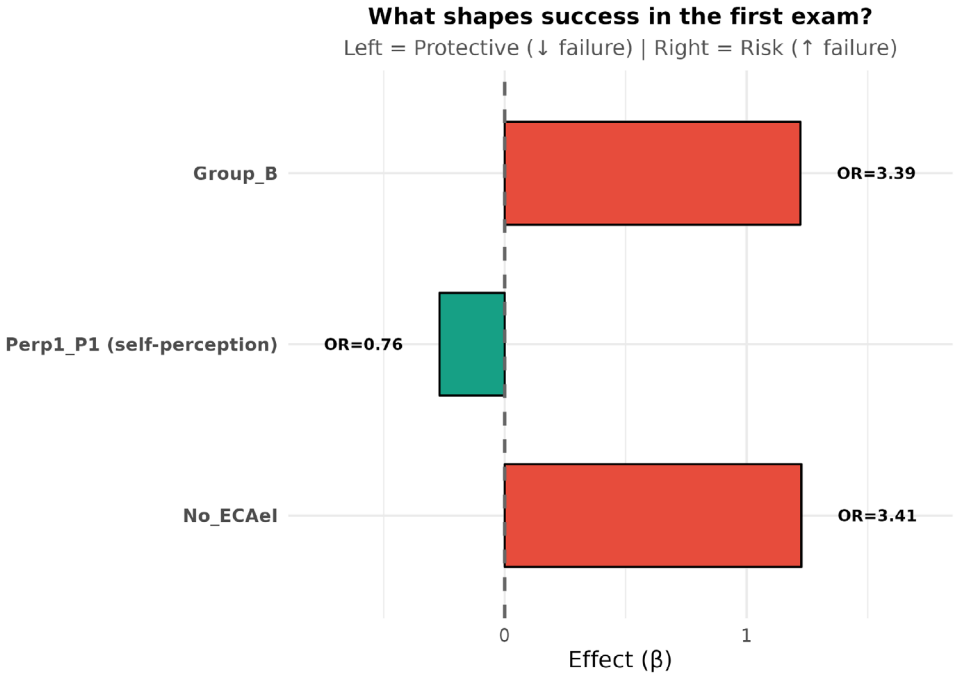


FIGURE 1 | Results of the logistic model for Unit 1. Bars represent the estimated coefficients and associated odds ratios. Negative values indicate protective factors, whereas positive values indicate risk factors.

magnitude of the association is highly imprecise and should be interpreted with caution.

Enrollment in programs outside ECAeI also remained a risk factor. In this unit, students from these programs were 3.16 times more likely to fail than their engineering counterparts (OR = 3.16, 95% CI: 1.12–8.96). Age was also positively associated with failure (OR = 1.26, 95% CI: 1.06–1.49), whereas sex showed a protective effect for the group coded as male (OR = 0.346, 95% CI: 0.129–0.930).

The self-perception variable related to the properties of random variables (Perp5_P2) showed a marginal protective effect. A 0.1-unit increase in self-perception was associated with an approximately 31.3% reduction in the odds of failure ($OR_{0.1} = 0.687$, 95% CI: 0.465–1.014). However, the confidence interval includes 1, so this result should be interpreted with caution. See Figure 2.

4.2.3 | Unit 3: Statistical Inference

In Unit 3, corresponding to Statistical Inference, students enrolled in programs outside ECAeI were 2.32 times more likely to fail than students from engineering programs (OR = 2.32, 95% CI: 1.01–5.32). Likewise, being at an early stage of the academic program was associated with higher odds of failure (OR = 2.58, 95% CI: 0.913–7.30), although this result should be interpreted as marginal.

In contrast to Unit 1, the variable Grupo_B showed no evidence of association with failure in this unit, suggesting that the section effect observed at the beginning of the course did not persist in subsequent units. See Figure 3.

4.2.4 | Unit 4: Linear Dependence Analysis

In Unit 4, the self-perception variable related to the ability to identify the elements involved in hypothesis testing for regression coefficients and to interpret the results in the context of the problem (*Perp2_P4*) was the strongest protective predictor in the model (OR = 0.001, 95% CI: 0.0000–0.183). Because this variable was rescaled to the $[0, 1]$ interval, the reported OR corresponds to a full one-unit increase on the rescaled scale, a magnitude that is not realistic for interpretative purposes. Therefore, it is more informative to interpret the effect for increments of 0.1 units. Under this parameterization, each one-tenth increase in student self-perception is associated with an approximate 47.9% reduction in the odds of failing the examination ($OR_{0.1} = e^{-6.524 \times 0.1} = 0.521$). This result suggests that greater confidence in the ability to conduct and interpret hypothesis tests in regression is associated with a substantial reduction in the risk of failure.

Examination duration also showed a moderate protective effect (OR = 0.971, 95% CI: 0.943–1.000), indicating that each additional minute spent on the assessment was associated with an approximate 2.9% reduction in the odds of failure.

Enrollment in programs outside ECAeI again emerged as a risk factor. In this unit, students from these programs were 2.98 times more likely to fail than engineering students (OR = 2.98, 95% CI: 1.14–7.82). See Figure 4.

4.3 | Cross-Unit Comparison

The most consistent pattern was the association between enrollment in a program outside ECAeI and higher odds of failure. This variable emerged as a risk factor in all four units. However, other

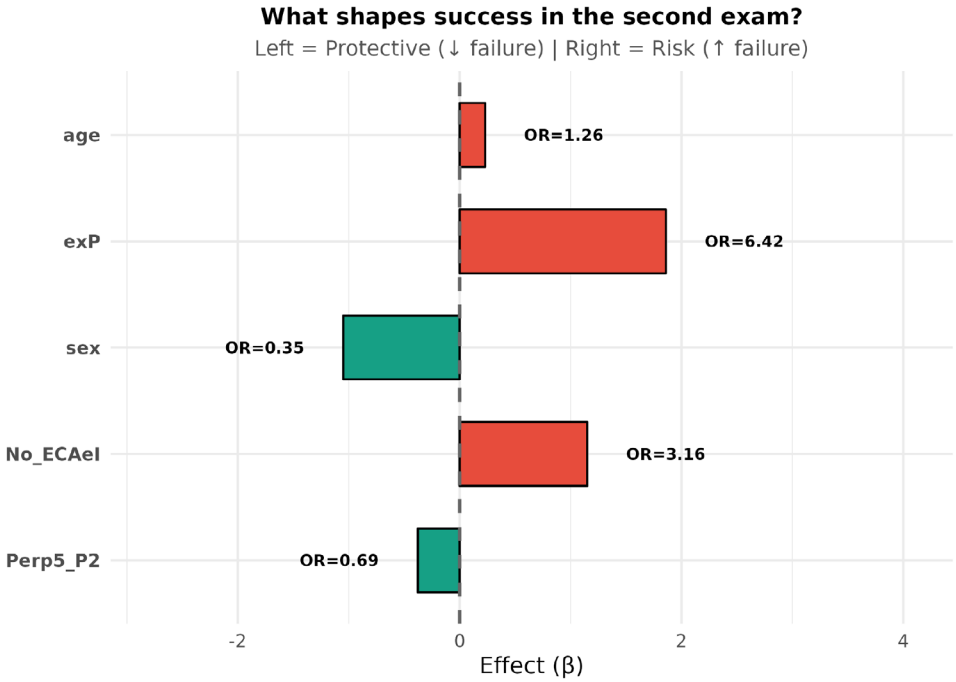


FIGURE 2 | Results of the logistic model for Unit 2. Bars represent the estimated coefficients and associated odds ratios. Negative values indicate protective factors, whereas positive values indicate risk factors. For the self-perception variable Perp5_P2, the displayed effect corresponds to a 0.1-unit increase on the rescaled $[0, 1]$ scale ($\beta_{0.1} = -0.375, \mathrm{OR}_{0.1} = 0.687$). Effects for all other predictors are shown on their original scales.

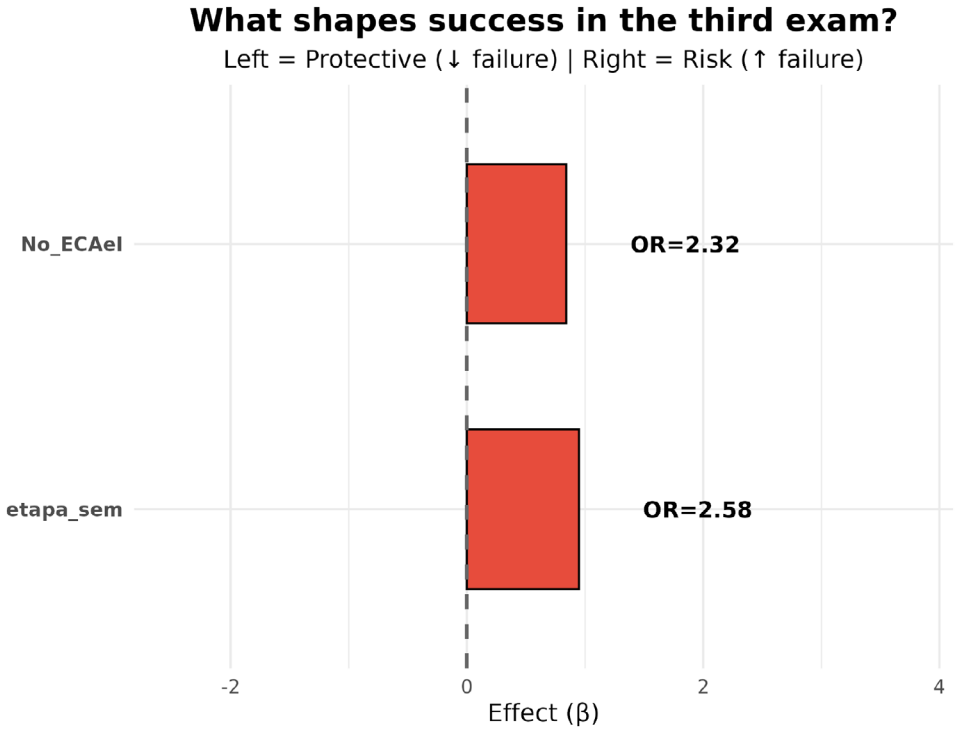


FIGURE 3 | Results of the logistic model for Unit 3. Bars represent the estimated coefficients and associated odds ratios. Negative values indicate protective factors, whereas positive values indicate risk factors.

predictors exhibited content-specific effects: prior programming experience was relevant in Unit 2, stage within the academic program in Unit 3, and self-perception regarding hypothesis testing for regression coefficients in Unit 4.

These findings indicate that academic vulnerability does not operate uniformly throughout the course. Rather, different factors become relevant depending on the content being assessed and the competencies required in each unit.

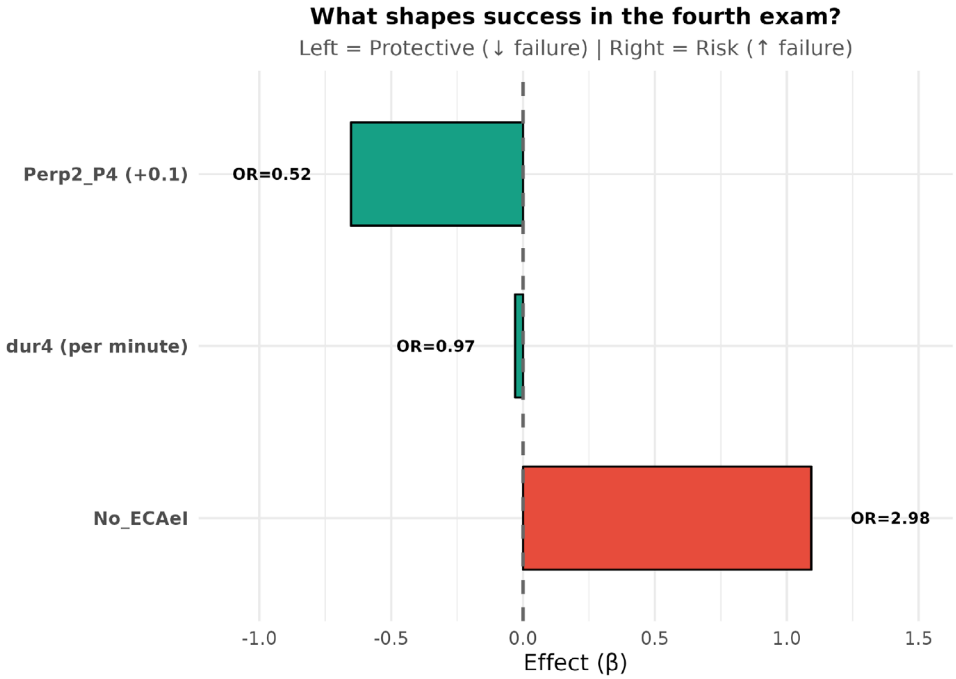


FIGURE 4 | Results of the logistic model for Unit 4. Bars represent the estimated coefficients and associated odds ratios. Negative values indicate protective factors, whereas positive values indicate risk factors.

4.4 | Robustness Results

To assess the sensitivity of the findings to the repeated-measures structure of the data, a logistic model with a student-level random intercept was estimated using the dataset in long format, consisting of 744 observations corresponding to 186 students observed across 4 units. The null model yielded an ICC of 0.073, indicating moderate within-student dependence. However, once the observed predictors were incorporated, the variance of the random intercept converged to zero, and the likelihood ratio test showed no improvement over a conventional logistic specification ($\chi^2(1) = 0.000$, $p = 0.500$).

A pooled model including Unit×Program interactions was also estimated. The inclusion of these interactions did not significantly improve model fit ($p = 0.711$), although this result should be interpreted cautiously given the number of additional parameters required.

Finally, we evaluated whether self-perception prior to each examination could be explained by performance on the preceding assessment. In the pooled model, a 0.1-unit increase in self-perception was associated with an approximately 15.6% reduction in the odds of failure ($OR_{0.1} = 0.844$, 95% CI: 0.768–0.928, $p < 0.001$), whereas failure on the previous examination did not reach statistical significance (OR = 1.402, 95% CI: 0.913–2.152, $p = 0.122$). These findings suggest that self-perception captures meaningful information about students' perceived preparedness and is not fully explained by performance on the preceding assessment.

Detailed estimates from the full models, the repeated-measures model, and the section-level robustness analyses are provided in the Supplementary Material.

4.5 | Robustness to Section-Level Clustering

Because the course was taught by three instructors across different sections, additional analyses were conducted to evaluate whether the observed results could be attributed to clustering effects associated with course section or instructor. To this end, the unit-specific models were re-estimated including an indicator variable for the section taught by the full-time faculty member (*Grupo_B*).

The results showed that the main predictors remained remarkably stable after the inclusion of this variable. For example, the effect associated with enrollment in a program outside ECAeI remained virtually unchanged across all four units: OR = 3.23 versus 3.41 in Unit 1, OR = 3.07 versus 3.16 in Unit 2, OR = 2.35 versus 2.32 in Unit 3, and OR = 2.88 versus 2.98 in Unit 4. Similarly, the effects associated with prior programming experience, stage within the academic program, examination duration, and self-perception variables exhibited only minimal changes after controlling for section.

The variable *Grupo_B* showed evidence of association only in Unit 1 (OR = 3.39, $p = 0.048$); however, this estimate should be interpreted with caution because of the small number of students in the section taught by the full-time faculty member ($n = 17$). In the remaining units, no statistically significant differences between sections were observed (Unit 2: OR = 1.42, $p = 0.628$; Unit 3: OR = 1.17, $p = 0.802$; Unit 4: OR = 2.26, $p = 0.189$). Taken together, these findings suggest that the main conclusions of the study do not depend substantially on the assigned section or instructor, providing additional evidence for the robustness of the unit-specific models.

5 | Discussion

The results of this study show that failure in an introductory statistics course is not a homogeneous phenomenon but rather varies substantially across thematic units. By disaggregating performance by content area, it was possible to identify differential patterns that would have remained hidden under a single aggregate measure of course performance. This finding supports the use of unit-level analyses as a more informative strategy for understanding learning difficulties and guiding targeted pedagogical interventions.

The findings can be interpreted through four complementary dimensions: contextual, cognitive, affective, and pedagogical factors.

From a contextual perspective, enrollment in programs outside the School of Applied Sciences and Engineering (ECAeI) was the most consistent predictor of failure across all four units analyzed. This result is consistent with previous research showing that students from non-STEM disciplines often enter statistics courses with weaker mathematical preparation, lower levels of confidence in their quantitative abilities, and more negative perceptions of statistics as a discipline (Gal and Ginsburg 1994; Rabin et al. 2021; Fabbricatore et al. 2023). The persistence of this effect throughout the course suggests that differences associated with disciplinary background constitute a structural source of academic vulnerability rather than a difficulty confined to specific topics.

From the cognitive and academic-preparation dimension, the results highlight the importance of prior preparation and students' stage within their academic trajectory. Students taking the course at earlier stages of their programs were more likely to fail inference-related content, suggesting that the progressive development of prior knowledge and academic experience may contribute to performance in conceptually demanding units.

From the affective dimension, perceived competence emerged as a relevant protective factor in specific units. These self-perception measures reflect students' beliefs about their ability to perform particular statistical tasks and are therefore closely related to self-efficacy. Although perceived competence concerns cognitively defined tasks, it is interpreted here primarily as an affective-motivational construct, while recognizing its connection with the cognitive dimension. This interpretation is consistent with previous research identifying self-efficacy and perceived cognitive competence as important predictors of success in statistics learning (Bandura 1997; Finney and Schraw 2003; Dempster and McCorry 2009; Rabin et al. 2021; Fabbricatore et al. 2023).

From a pedagogical and curricular perspective, the differences observed across units reinforce the idea that students do not experience the statistics curriculum as a uniform progression of skills. Consistent with Garfield and Ben-Zvi (2007) and Garfield et al. (2008), concepts related to variability, inference, and hypothesis testing generated substantially greater difficulties than more procedural tasks. These findings support recommendations to promote instructional approaches centered on statistical reasoning, conceptual integration, and the progressive scaffolding of complex ideas, rather than focusing exclusively on the mastery of procedures and algorithms (Olani et al. 2011; Garfield and Ben-Zvi 2007; Garfield et al. 2008).

Finally, the robustness analyses provide additional support for these conclusions. The repeated-measures and section-level analyses indicated that the main patterns remained stable after accounting for within-student dependence, potential instructor-related differences, and prior performance. However, these analyses do not eliminate possible selection associated with course persistence or complete-case inclusion. Accordingly, the cross-unit patterns should be interpreted conditional on the analytical sample.

In summary, the findings support a shift from global performance indicators toward learning analytics approaches at the thematic-unit level. This perspective makes it possible to identify more precisely the conceptual bottlenecks faced by students and provides more useful information for designing timely and targeted pedagogical interventions throughout the semester.

6 | Conclusions and Recommendations

This study demonstrates that failure in an introductory statistics course is not distributed uniformly across the different components of the curriculum. By analyzing performance at the thematic-unit level, substantial differences were identified both in failure rates and in the factors associated with the risk of academic failure—patterns that would have remained hidden under a single aggregate measure of course performance. In particular, the units on Descriptive Statistics and Statistical Inference emerged as the most challenging components for students.

Consistently, students enrolled in programs outside the School of Applied Sciences and Engineering (ECAeI) exhibited higher odds of failure across all four units analyzed. In contrast, variables such as prior programming experience, stage within the academic program, and perceived competence showed content-specific effects. These findings highlight the importance of understanding statistical learning as a multidimensional process in which different factors become relevant depending on the skills and knowledge required in each unit.

From a pedagogical perspective, the findings support the use of learning analytics at the thematic-unit level to identify conceptual difficulties and groups of students at risk of academic failure at an early stage. Beyond final course grades, the systematic monitoring of performance in specific content areas can provide more useful information for designing timely and targeted interventions.

This study also highlights several opportunities for future research. First, it would be desirable to incorporate additional variables related to class attendance, motivation, students' interest in the subject, and study strategies, all of which have been identified in the literature as potentially relevant factors in explaining academic performance. Second, it is particularly important to examine the practical component of the course conducted in R laboratories, which accounts for the remaining

40% of the final course grade and may provide valuable insights into the development of computational and data-analysis skills. Finally, longitudinal studies that follow students throughout their university trajectory would allow for a better understanding of how statistical reasoning evolves over time and how early difficulties influence subsequent learning.

Taken together, the findings suggest that understanding failure in statistics requires moving beyond aggregate performance indicators toward a more detailed evaluation of how students engage with different statistical concepts throughout their learning process.

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Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in repository.eafit.edu.co at https://repository.eafit.edu.co/entities/publication/776b49b5-d51d-4ea5-8c14-b40c3d3fbe81.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Full logistic regression results—Unit 1 (descriptive statistics). **Table S2:** Full logistic regression results—Unit 2 (random phenomena). **Table S3:** Full logistic regression results—Unit 3 (statistical inference). **Table S4:** Full logistic regression results—Unit 4 (linear regression). **Table S5:** Variance decomposition

and model comparison statistics for the repeated-measures specification. **Table S6:** Fixed-effect estimates from the pooled repeated-measures logistic model. **Table S7:** Likelihood ratio test comparing pooled repeated-measures specifications. **Table S8:** Unit 1 robustness assessment including section membership (Grupo_B). **Table S9:** Unit 2 robustness assessment including section membership (Grupo_B). **Table S10:** Unit 3 robustness assessment including section membership (Grupo_B). **Table S11:** Unit 4 robustness assessment including section membership (Grupo_B). **Table S12:** General characteristics of the complete-case analytical sample ($N = 186$). **Table S13:** Course units and learning outcomes. **Table S14:** Perception survey items by unit.